

Account Lockout Detection and Investigation using Windows Security Logs and Splunk

Executive Summary

This investigation documents the detection and analysis of a Windows account lockout event using Microsoft Windows Security Logs and Splunk Enterprise within a controlled SOC detection engineering lab.

A controlled attack simulation generated multiple failed interactive authentication attempts against the local administrator account. After five consecutive authentication failures, Windows automatically locked the account according to the configured Account Lockout Policy.

The investigation reconstructed the complete attack timeline using Windows Security Event IDs 4625 (Failed Logon) and 4740 (Account Lockout). Splunk Enterprise was used to search, correlate, visualize, and investigate the authentication activity.

Although the activity was intentionally generated in a laboratory environment, the behavioral pattern closely resembles credential guessing, brute-force authentication attempts, and password spraying attacks commonly investigated by enterprise SOC analysts.

Objective

The objective of this investigation was to detect, investigate, and reconstruct an account lockout event using Windows Security Logs and Splunk.

The investigation focused on identifying repeated authentication failures, correlating account lockout events, reconstructing the authentication timeline, validating Windows account lockout policies, and developing practical SPL searches for authentication monitoring.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Brute Force	T1110
Initial Access	Valid Accounts	T1078

Lab Environment

Component	Technology
SIEM	Splunk Enterprise 10.4
Endpoint	Windows 11 Home
Log Source	Windows Security Log
Forwarder	Splunk Universal Forwarder
Index	windows

Account Lockout Policy

The following Windows Account Lockout Policy was configured before the simulation.

Setting	Value
Lockout Threshold	5 Failed Attempts
Lockout Duration	5 Minutes
Observation Window	5 Minutes

Attack Simulation

The attack simulation was performed directly from the Windows lock screen. The existing local administrator account (**scoadmin**) was selected and an incorrect password was intentionally entered five consecutive times. After the fifth failed authentication attempt, Windows enforced the configured Account Lockout Policy and locked the account. This simulation reproduces authentication behavior commonly observed during:

- Credential guessing
- Interactive brute-force attacks
- Password spraying (when multiple users are targeted)
- Unauthorized access attempts

Telemetry Sources

Log Source	Event ID	Purpose
Windows Security Log	4625	Failed Interactive Logon
Windows Security Log	4740	Account Lockout

Attack Timeline

Time	Event	Description
09:04:00	4625	Failed authentication attempt
09:04:02	4625	Failed authentication attempt
09:04:05	4625	Failed authentication attempt
09:04:07	4625	Failed authentication attempt
09:04:10	4625	Fifth failed authentication
09:04:10	4740	Windows account locked

Splunk Investigation

Query 1 — Failed Authentication Events

```
index=windows EventCode=4625
```

Purpose: Retrieve all failed authentication events recorded by Windows Security Logs.

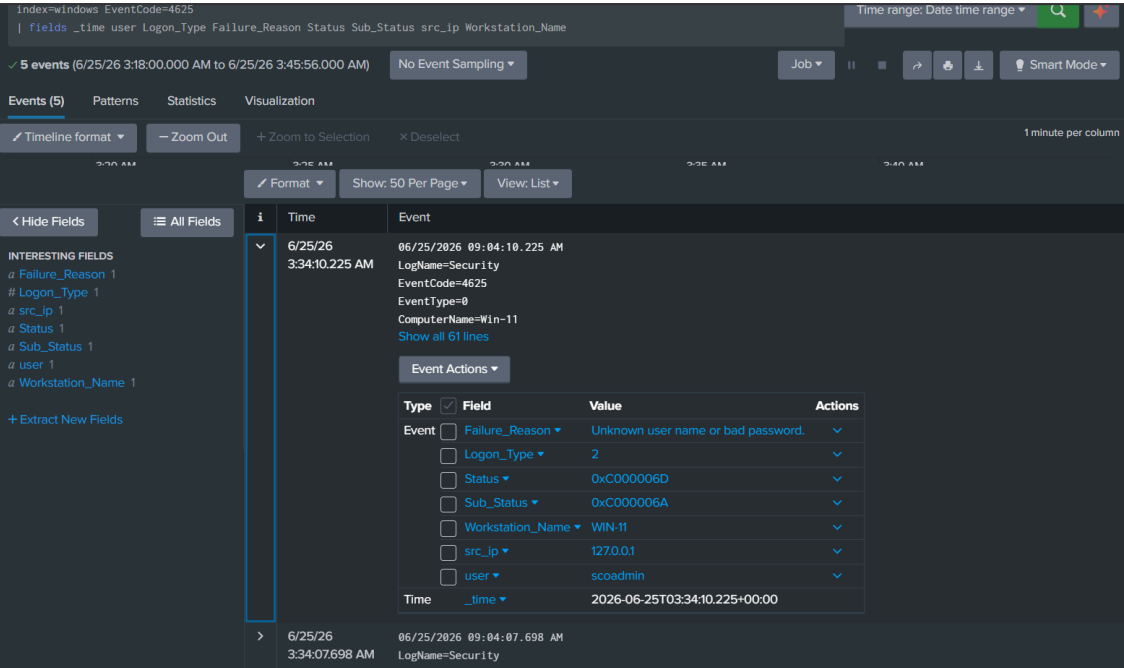


Figure 1 - Splunk search results showing Windows Security Event ID 4625 failed authentication

Query 2 — Account Lockout Event

```
index=windows EventCode=4740
```

Purpose:Identify Windows account lockout events generated after repeated authentication failures.

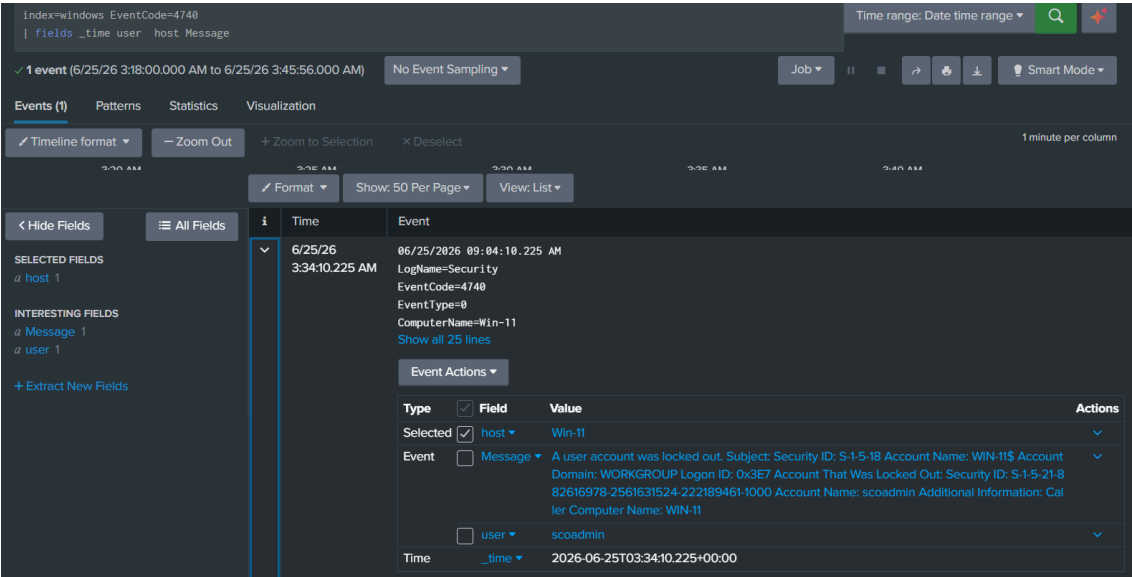


Figure 2 — Splunk search results showing Windows Security Event ID 4740 account lockout event.

Query 3 — Authentication Timeline Reconstruction

```
index=windows (EventCode=4625 OR EventCode=4740) user="scoadmin" | sort _time | table _time EventCode user src_ip Workstation_Name Failure_Reason Status Sub_Status
```

Purpose: Reconstruct the complete authentication timeline leading to the account lockout.

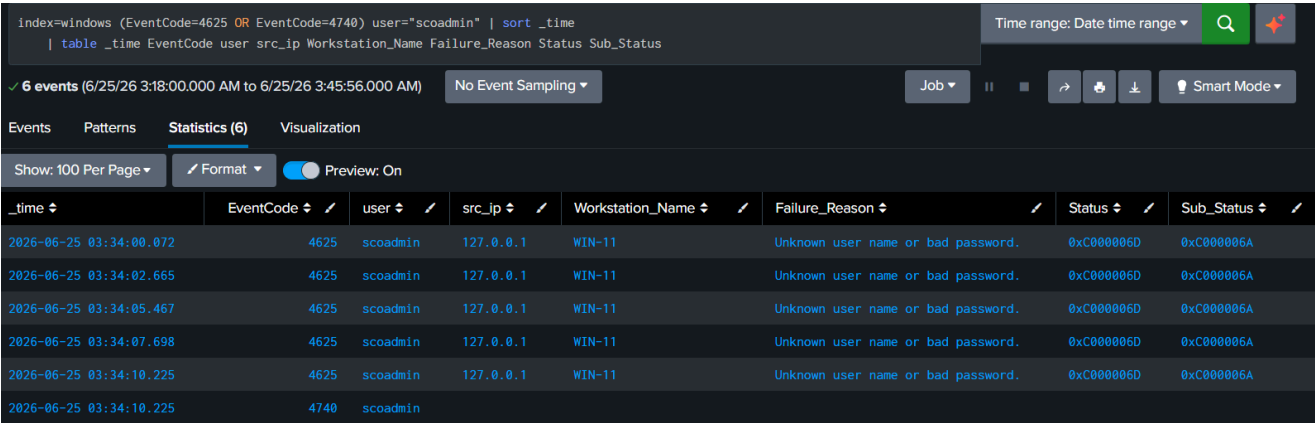


Figure 3 — Authentication timeline reconstructed in Splunk showing five consecutive failed logon attempts followed by automatic account lockout.

Query 4 — Detection Logic

```
index=windows (EventCode=4625 OR EventCode=4740) | bin _time span=5m | stats
count(eval(EventCode=4625)) as FailedLogons count(eval(EventCode=4740)) as
Lockouts values(src_ip) as SourceIP values(user) as TargetUser
earliest(_time) as FirstSeen latest(_time) as LastSeen by host _time | where
FailedLogons>=5
```

Purpose: Detect accounts that experience five or more failed authentication attempts within a five-minute window, followed by account lockout activity.

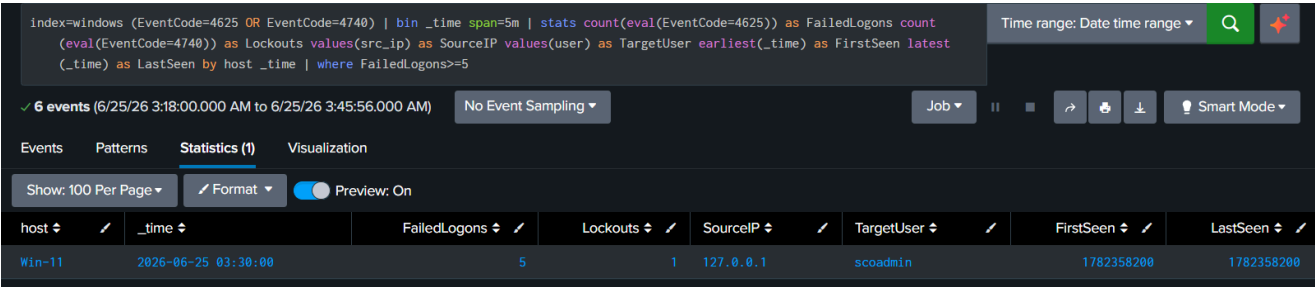


Figure 4 — Detection query correlating failed logon attempts and account lockout events for rapid analyst triage.

Investigation Findings

Windows Event ID 4625 – Failed Authentication

Splunk identified five consecutive Windows Security Event ID 4625 events corresponding to failed interactive logon attempts against the **scoadmin** account.

Key Findings

Field	Value
Event ID	4625
Target User	scoadmin
Logon Type	2 (Interactive)
Authentication Package	Negotiate
Failure Reason	Unknown user name or bad password
Status	0xC000006D
SubStatus	0xC000006A

Field	Value
Source Address	127.0.0.1
Workstation	WIN-11



Figure 2 — Windows Security Event ID 4625 recording failed interactive authentication against the local administrator account.

Analyst Assessment

The repeated failures occurred within a short time window, matching the configured lockout threshold. While the source address (**127.0.0.1**) indicates the activity originated locally in the lab, the behavioral pattern closely resembles brute-force or credential guessing attacks that SOC analysts routinely investigate.

Windows Event ID 4740 – Account Lockout

Following the fifth failed authentication attempt, Windows generated Security Event ID 4740 indicating that the account had been locked.

Key Findings

Field	Value
Event ID	4740
Locked Account	scoadmin
Caller Computer	WIN-11

Field	Value
Action	Account Locked

Security Number of events: 177 (1) New events available

Keywords	Date and Time	Source	Event...	Task Category
Audit Succ...	6/25/2026 9:04:10...	Microsoft Wi...	4740	User Account ...

Event 4740, Microsoft Windows security auditing.

GeneralDetails

A user account was locked out.

Subject:

Security ID:SYSTEM
Account Name:WIN-11\$
Account Domain:WORKGROUP
Logon ID:0x3E7

Account That Was Locked Out:

Security ID:WIN-11\scoadmin
Account Name:scoadmin

Additional Information:

Caller Computer Name:WIN-11

Log Name:Security

Source:Microsoft Windows security

Event ID:4740

Level:Information

User:N/A

OpCode:Info

Logged:6/25/2026 9:04:10 AM

Task Category:User Account Manager

Keywords:Audit Success

Computer:Win-11

Analyst Assessment

The account lockout occurred immediately after the fifth failed authentication attempt, confirming that the configured Account Lockout Policy functioned as expected. Correlating Event IDs 4625 and 4740 provides a clear sequence of authentication failures followed by enforcement of the lockout policy.

IOC Summary

IOC	Value
Target Account	scoadmin
Host	WIN-11
Source IP	127.0.0.1
Authentication Package	Negotiate
Logon Type	2
Event IDs	4625, 4740
Status	0xC000006D
SubStatus	0xC000006A

Detection Logic

The detection logic is based on correlating multiple failed authentication events with a subsequent account lockout event for the same user within a short period.

Example Detection SPL

```
index=windows (EventCode=4625 OR EventCode=4740)
| stats count values(EventCode) as Events earliest(_time) as FirstSeen
latest(_time) as LastSeen by user
| where count>=5
```

False Positive Considerations

Repeated authentication failures are not always malicious. Possible benign causes include:

- Users entering incorrect passwords repeatedly.
- Cached credentials after a password change.
- Misconfigured scheduled tasks or services.
- Applications using outdated credentials.
- Automated scripts attempting authentication with expired passwords.

Analysts should correlate account lockouts with user activity, source systems, authentication patterns, and surrounding security events before concluding malicious activity.

Detection Recommendations

- Alert on repeated Event ID 4625 occurrences targeting the same account within a defined time window.
- Correlate Event ID 4740 with preceding authentication failures to identify the source of the lockout.
- Prioritize investigations involving privileged accounts or multiple lockouts across different systems.
- Monitor authentication attempts originating from external or unusual network locations.
- Review authentication trends to identify recurring brute-force or credential guessing activity.

Conclusion

This investigation successfully demonstrated the detection and reconstruction of a Windows account lockout event using Windows Security Logs and Splunk Enterprise. Five

consecutive failed interactive logon attempts against the **scoadmin** account triggered Windows Security Event ID 4740, confirming enforcement of the configured Account Lockout Policy. By correlating Event IDs **4625** and **4740**, the investigation produced a complete authentication timeline representative of credential-based attacks frequently investigated by SOC analysts. The project reinforces practical skills in Windows authentication analysis, event correlation, SPL-based threat hunting, and detection engineering for authentication monitoring.